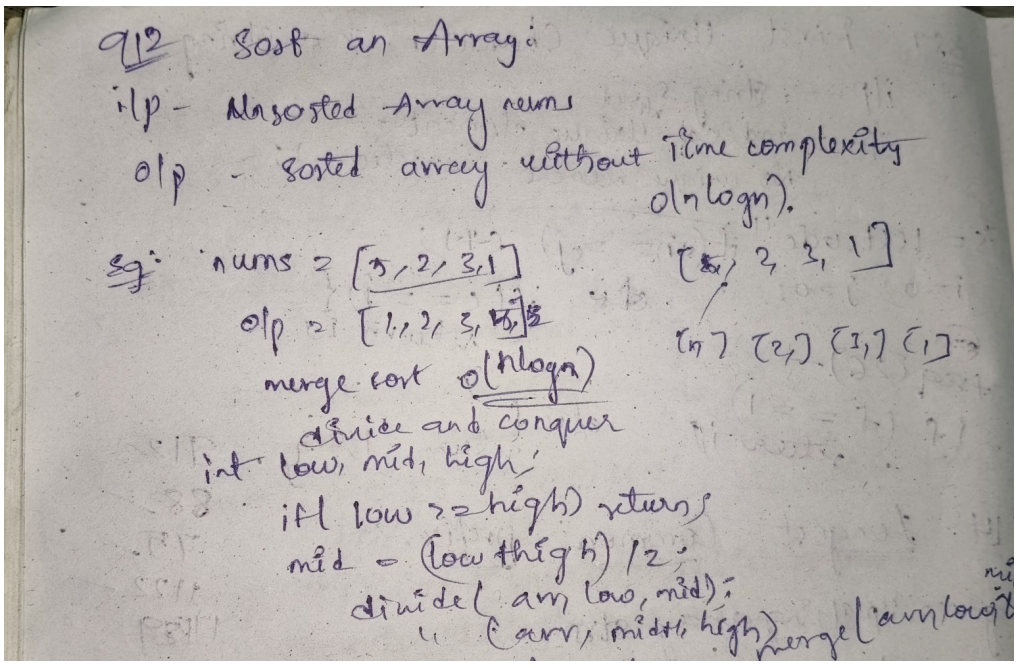
K. SANJANA
2505A41L63

11. Sort an Array

```java
class Solution {
    void merge(int arr[], int low, int mid, int high) {
        int temp[] = new int[high - low + 1];
        int i = low;
        int j = mid + 1;
        int k = 0;
        while(i <= mid && j <= high) {
            if(arr[i] < arr[j]) {
                temp[k] = arr[i];
                i++;
            }
            else {
                temp[k] = arr[j];
                j++;
            }
            k++;
        }
        while(i <= mid) {
            temp[k] = arr[i];
            i++;
            k++;
        }
        while(i <= high) {
```



11. Sort an Array (continued)

Code

va Auto

```
            i++;
            k++;
        }
        while(j <= high) {
            temp[k] = arr[j];
            j++;
            k++;
        }
        for(i = low; i <= high; i++) {
            arr[i] = temp[i - low];
        }
    }
    void divide(int arr[], int low, int high) {
        if(low >= high) return;
        int mid = (low + high) / 2;
        divide(arr, low, mid);
        divide(arr, mid + 1, high);
        merge(arr, low, mid, high);
    }
    public int[] sortArray(int[] nums) {
        divide(nums, 0, nums.length - 1);
        return nums;
    }
}
```

LeetCode Link: https://leetcode.com/problems/sort-an-array/

12. Merge Sorted Array

https://leetcode.com/problems/merge-sorted-array/description/

Problem List

Code

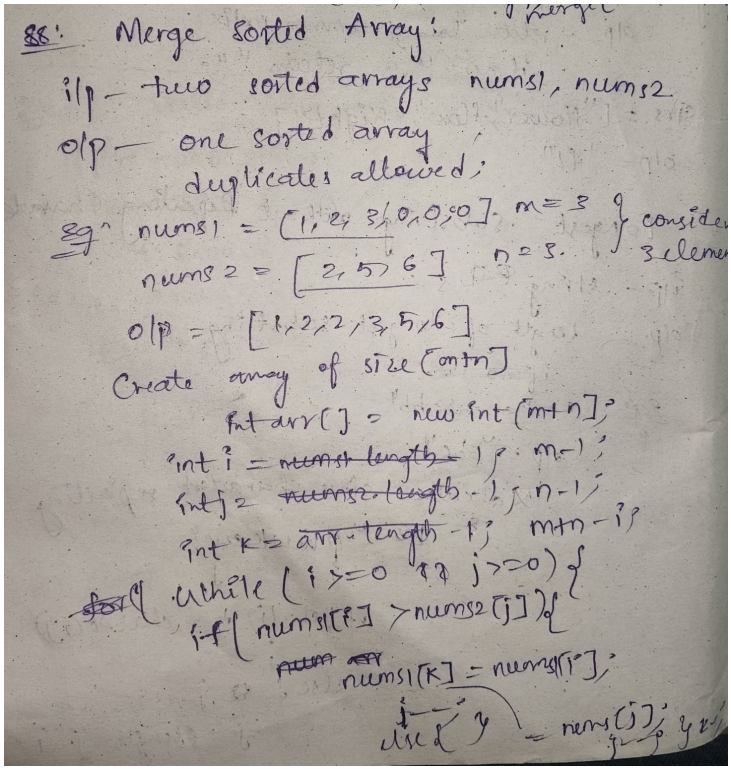
Auto

```java
class Solution {
    public void merge(int[] nums1, int m, int[] nums2, int n) {

        int i = m - 1;
        int j = n - 1;
        int k = m + n - 1;

        while (i >= 0 && j >= 0) {
            if (nums1[i] > nums2[j]) {
                nums1[k] = nums1[i];
                i--;
            } else {
                nums1[k] = nums2[j];
                j--;
            }
            k--;
        }

        while (j >= 0) {
            nums1[k] = nums2[j];
            j--;
            k--;
        }
    }
}
```



LeetCode Link: https://leetcode.com/problems/merge-sorted-array/

13. Sort Colors

Problem List

Code

Java Auto

```java
class Solution {
    public void sortColors(int[] nums) {

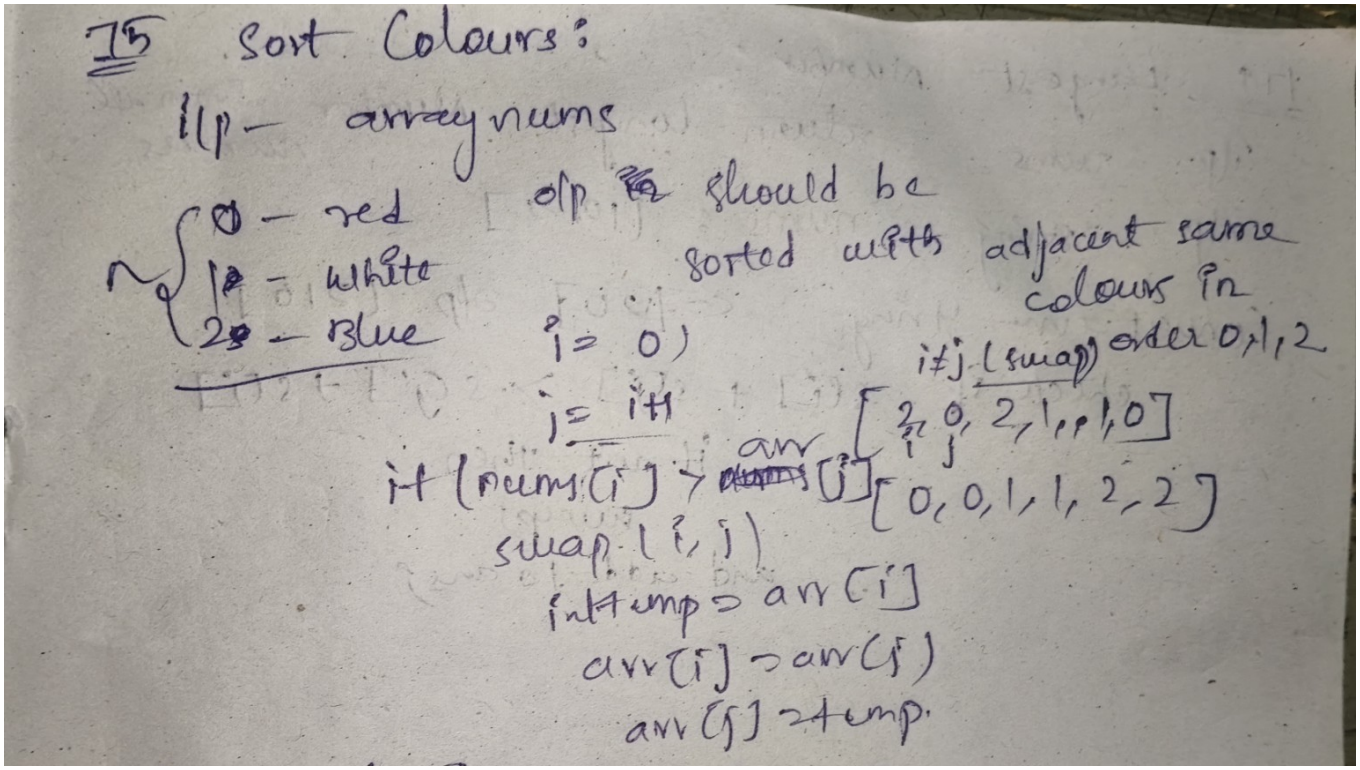
        for(int i = 0; i < nums.length; i++) {

            for(int j = i + 1; j < nums.length; j++) {

                if(nums[i] > nums[j]) {

                    int temp = nums[i];
                    nums[i] = nums[j];
                    nums[j] = temp;

                }
            }
        }
    }
}
```



LeetCode Link: https://leetcode.com/problems/sort-colors/

14. Relative Sort Array

Problem List

Code

Auto

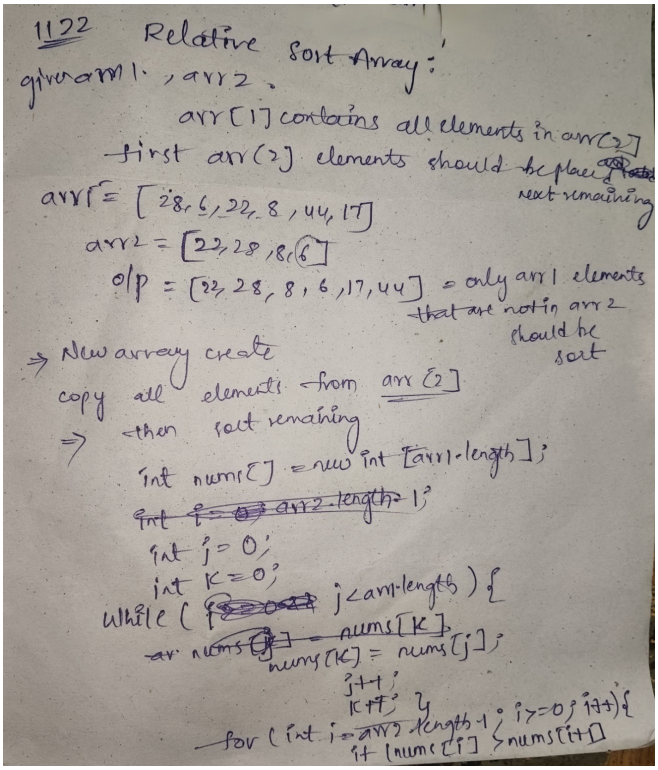
```java
class Solution {
    public int[] relativeSortArray(int[] arr1, int[] arr2) {

        int nums[] = new int[arr1.length];
        int k = 0
        for(int j = 0; j < arr2.length; j++) {
            for(int i = 0; i < arr1.length; i++) {
                if(arr1[i] == arr2[j]) {
                    nums[k] = arr1[i];
                    k++;
                    arr1[i] = -1;
                }
            }
        }
        int start = k;
        for(int i = 0; i < arr1.length; i++) {
            if(arr1[i] != -1) {
                nums[k] = arr1[i];
                k++;
            }
        }
        for(int i = start; i < nums.length; i++) {
            for(int j = i + 1; j < nums.length; j++) {
                if(nums[i] > nums[j]) {
                    int temp = nums[i];
                    nums[i] = nums[j];
                    nums[j] = temp;
                }
            }
        }
        return nums;
```

aved

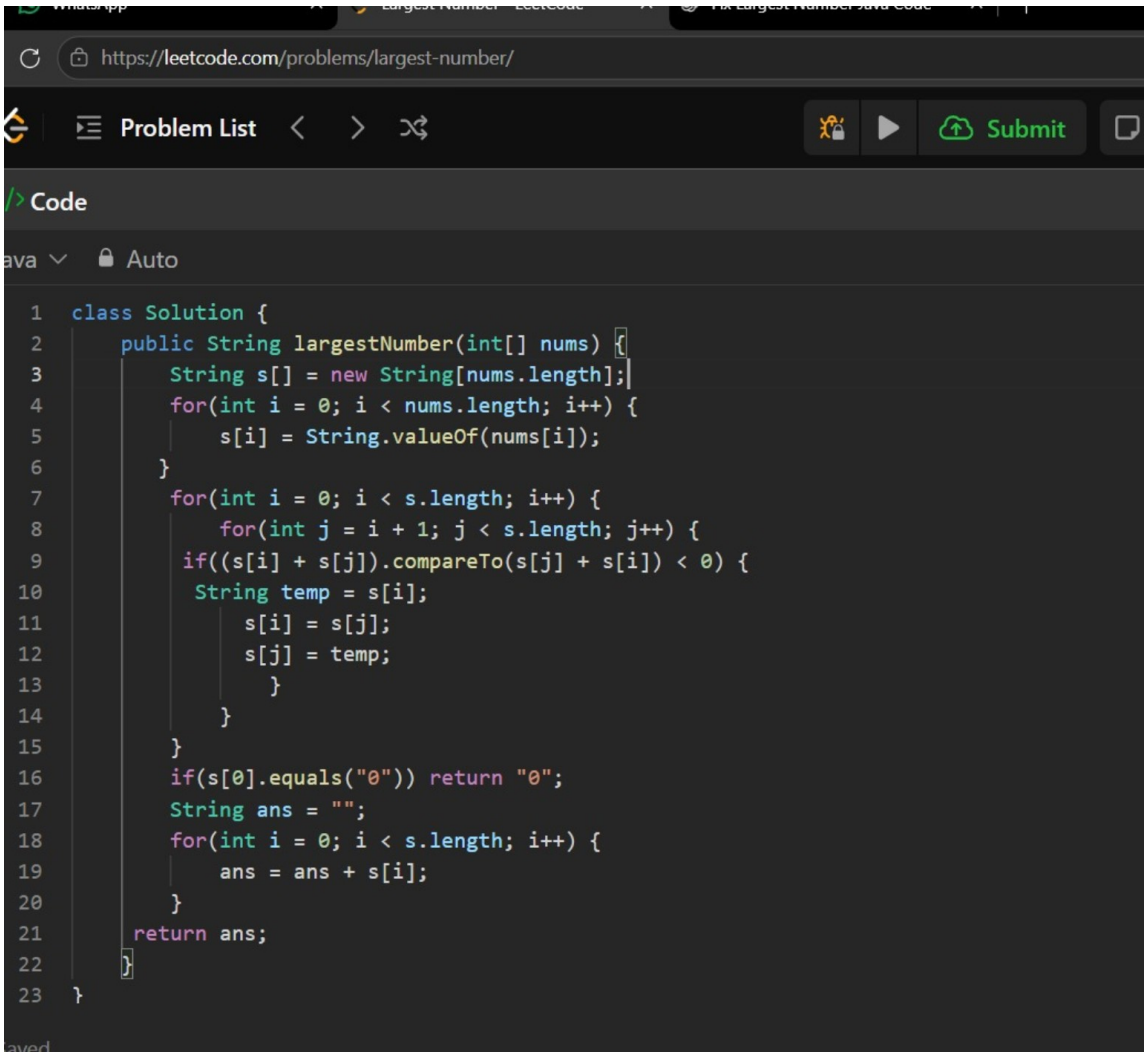
25°C
Mostly cloudy

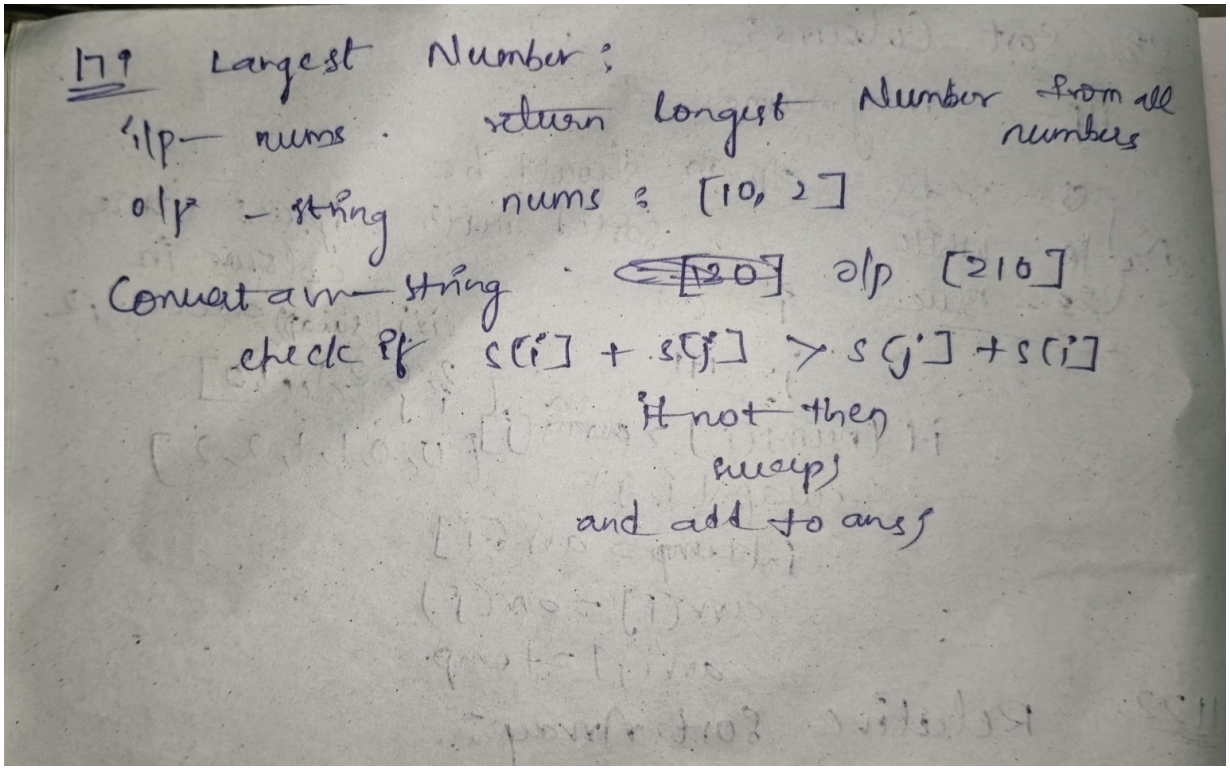
Search



LeetCode Link: https://leetcode.com/problems/relative-sort-array/description/

15. Largest Number





LeetCode Link: https://leetcode.com/problems/largest-number/